Nucleus: The protons and neutrons [collectively called **nucleons**]are found in the nucleus of an atom.

Orbits: Orbit is defined as the path, by which electrons revolve around the nucleus.

DO YOU KNOW?

Besides the fundamental particles like protons, electrons and neutrons some more particles are discovered in the nucleus of an atom. They include mesons, neutrino, antineutrino, positrons etc.

11.5 Atomic number and Mass number

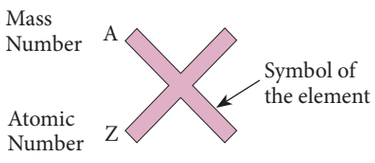
Only hydrogen atoms have one proton in their nuclei. Only helium atoms have two protons. Indeed, only gold atoms have 79 protons. This shows that the number of protons in the nucleus of an atom decides which element it is. This very important number is known as the **atomic number** (proton number, given the symbol Z) of an atom.

Atomic number(Z) = Number of protons = Number of electrons

Protons alone do not make up all of the mass of an atom. The neutrons in the nucleus also contribute to the total mass. The mass of the electron can be regarded as so small that it can be ignored. As a proton and a neutron have the same mass, the mass of a particular atom depends on the total number of protons and neutrons present. This number is called the **mass number** (or nucleon number, given the symbol A) of an atom.

Mass number = Number of protons + Number of neutrons

For any element, the atomic numbers are shown as subscripts and mass number are shown as superscripts.



For example, nitrogen is written as $^{14}_{7}N$

Here **7** is its atomic number and **14** is its mass number.

Activity 1

Symbolically represent the following atoms using atomic number and mass number. [Refer table 11.1]

a) Carbon b) Oxygen
c) Silicon d) Beryllium

The difference between the mass number of an element and its atomic number gives the number of neutrons present in one atom of the element.

Number of neutrons (n) = Mass number (A) – Atomic number (z)

For example, the number of neutrons in one atom of $^{24}_{12}Mg$ is

Number of neutrons (n) = 24 (A) – 12 (Z) = 12

Test Yourself

Calculate the number of neutrons in the following atoms:

a) $^{27}_{13}Al$ b) $^{31}_{15}P$ c) $^{190}_{76}Os$ d) $^{54}_{24}Cr$

DO YOU KNOW?

Atomic number is designated as Z why?
Z stands for Zahl, which means NUMBER in German.
Z can be called Atomzahl or atomic number
A is the symbol recommened in the ACS style guide instead of M (massenzahl in German).

Problem 1

Calculate the atomic number of an element whose mass number is 39 and number of neutrons is 20. Also find the name of the element.

Solution

Mass Number = Atomic number + Number of neutrons
Atomic Number = Mass number – Number of neutrons
= 39 – 20
Atomic Number = 19

Element having atomic number 19 is Potassium (K)

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